

Grade 8
Unit 1
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. The Zero Exponent Law states $(x)^0 = 1$

$a^0 = 1$ is the zero exponent law

2. The Power of a Quotient Exponent Law states $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$

Apply the exponent to all values inside the parenthesis.

3. The Quotient of Powers Exponent Law states $\frac{x^m}{x^n} = x^{m-n}$

When bases are the same, subtract the exponents.

For questions 4-7, rewrite each expression as an equivalent exponential expression.

4. $\frac{\left(\frac{4}{5}\right)^{18}}{\left(\frac{4}{5}\right)^{18}} = \left(\frac{4}{5}\right)^{18-18} \rightarrow \left(\frac{4}{5}\right)^0 = 1$

5. $\frac{(-4.3)^9}{(-4.3)^4} = (-4.3)^{9-4} = (-4.3)^5$

6. $\left(\frac{13}{7}\right)^6 = \frac{13^6}{7^6}$

7. $(7^4)^{10} = 7^{4 \cdot 10} = 7^{40}$

For questions 8-10, rewrite each expression as an equivalent expression, then identify which exponent law that was applied.

8. $(6.5)^4 \cdot (6.5)^7 =$

$(6.5)^{4+7} = (6.5)^{11}$

- ☒ Product of powers
- ☐ Quotient of powers
- ☐ Power of a product
- ☐ Power of a quotient
- ☐ Power of a power
- ☐ Zero exponent

9. $\frac{8^9}{8} = 8^{9-1} = 8^8$

- ☐ Product of powers
- ☒ Quotient of powers
- ☐ Power of a product
- ☐ Power of a quotient
- ☐ Power of a power
- ☐ Zero exponent

10. $(-9.75)^0 = = 1$

- ☐ Product of powers
- ☐ Quotient of powers
- ☐ Power of a product
- ☐ Power of a quotient
- ☐ Power of a power
- ☒ Zero exponent